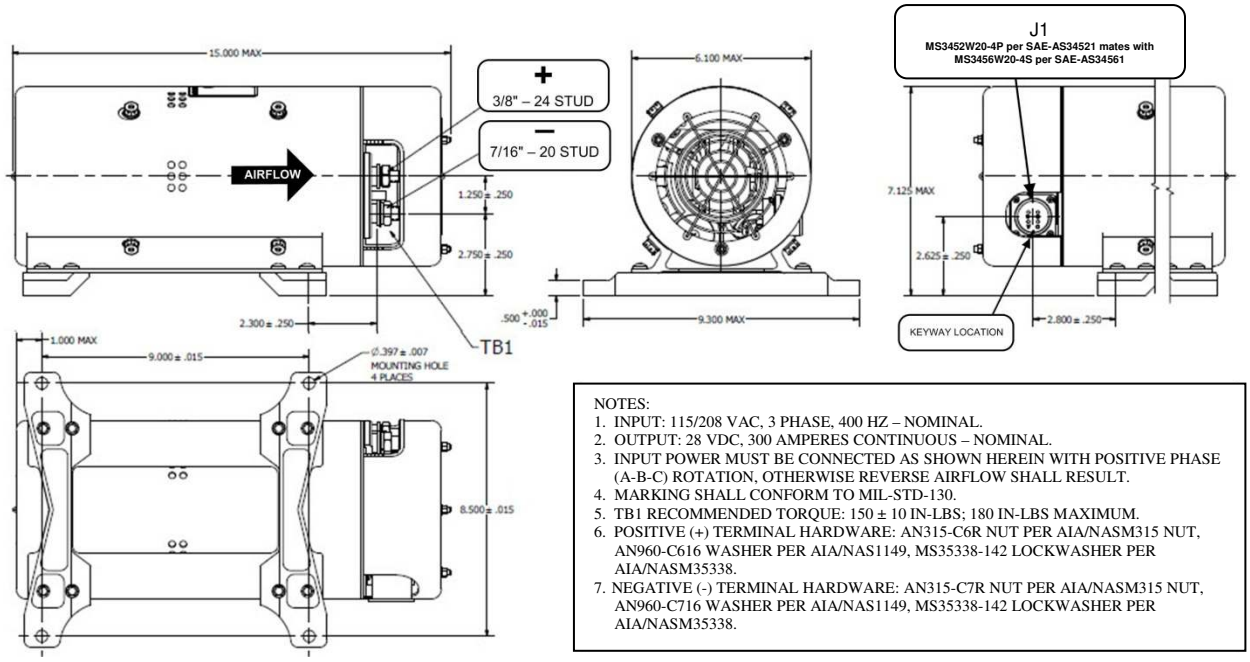
INCH-POUND
MIL-PRF-7115/4A(AS)
21 August 2014
SUPERSEDING
MIL-PRF-7115/4(AS)
11 July 2012

PERFORMANCE SPECIFICATION SHEET

CONVERTER, AIRCRAFT
300 AMPERE, TYPE II, CLASS A, FAN COOLED

This specification is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for the product described herein shall consist of this specification sheet and MIL-PRF-7115.



NOTES:
1. INPUT: 115/208 VAC, 3 PHASE, 400 HZ – NOMINAL.
2. OUTPUT: 28 VDC, 300 AMPERES CONTINUOUS – NOMINAL.
3. INPUT POWER MUST BE CONNECTED AS SHOWN HEREIN WITH POSITIVE PHASE (A-B-C) ROTATION, OTHERWISE REVERSE AIRFLOW SHALL RESULT.
4. MARKING SHALL CONFORM TO MIL-STD-130.
5. TB1 RECOMMENDED TORQUE: 150 ± 10 IN-LBS; 180 IN-LBS MAXIMUM.
6. POSITIVE (+) TERMINAL HARDWARE: AN315-C6R NUT PER AIA/NASM315 NUT, AN960-C616 WASHER PER AIA/NAS1149, MS35338-142 LOCKWASHER PER AIA/NASM35338.
7. NEGATIVE (-) TERMINAL HARDWARE: AN315-C7R NUT PER AIA/NASM315 NUT, AN960-C716 WASHER PER AIA/NAS1149, MS35338-142 LOCKWASHER PER AIA/NASM35338.

Weight: 28.0 pounds maximum

FIGURE 1. Dimensions and configuration for P/N M7115/04.

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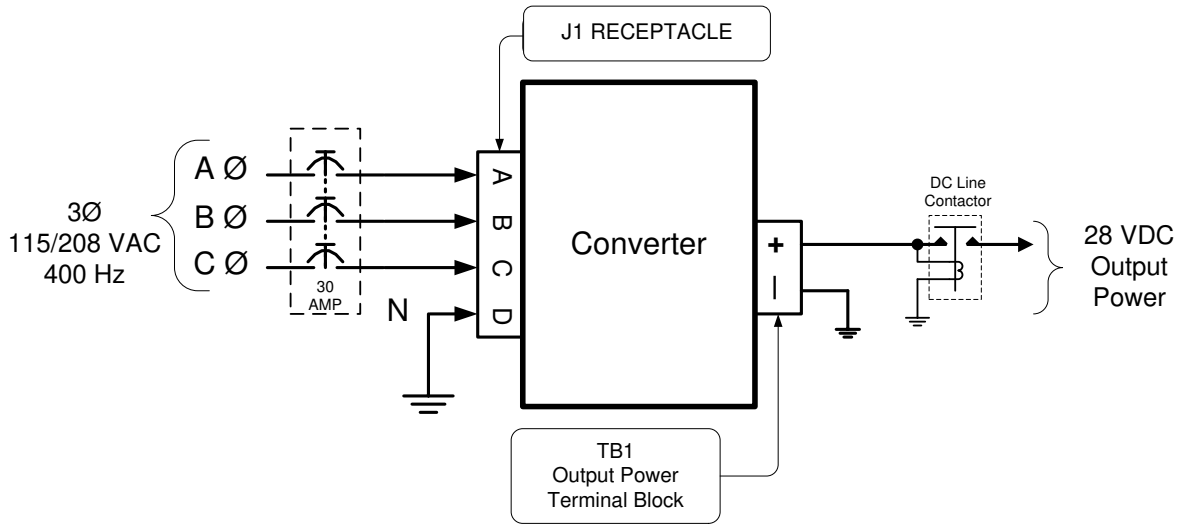


FIGURE 2. Connection diagram.

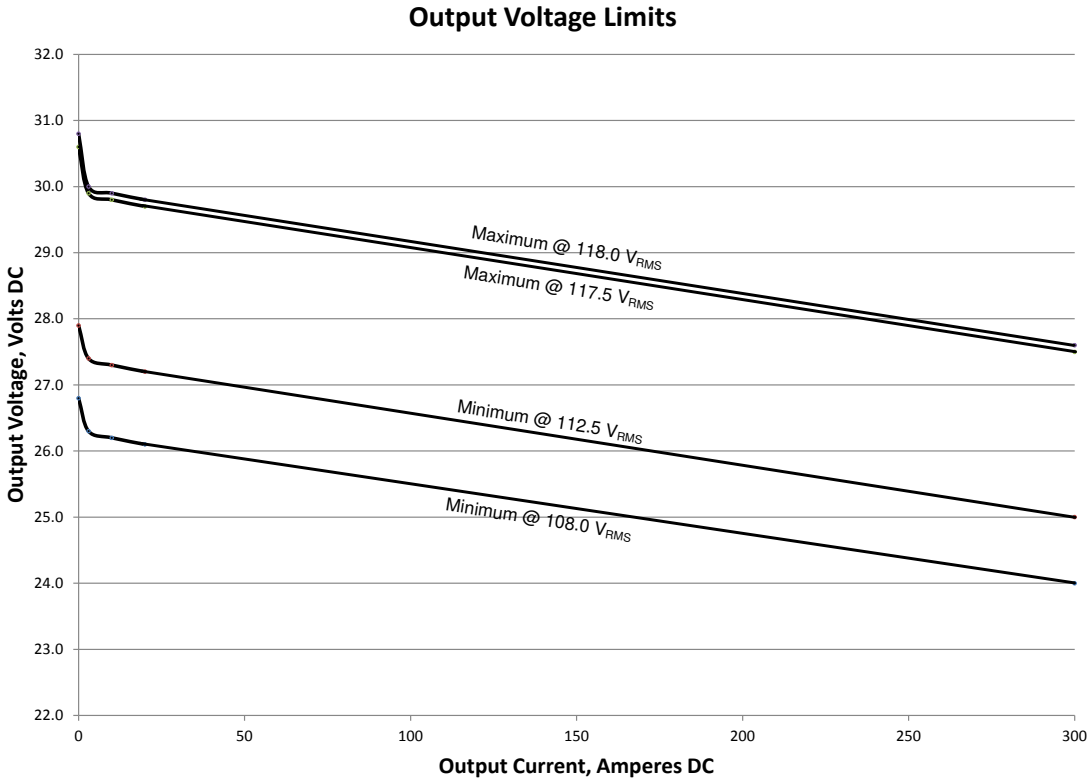
TABLE I. Converter characteristics.

AC INPUT:	
Voltage Range	112.5 - 117.5
Input Current, per phase	27 amperes (@300 A output)
Frequency Range	380 - 420 Hertz
DC Output:	
Voltage	28 Volts (Nominal)
Maximum Continuous Load Current	300 Amperes

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TABLE II. Acoustical noise sound levels.

Octave Band HZ	Sound Pressure Level dB (Ref -0.0002 Microbar)
31	91
63	92
125	93
250	94
500	90
1000	87
2000	87
4000	77



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Input (VRMS)	108.0	112.5	117.5	118.0
Output (Amps DC)	Minimum VDC	Minimum VDC	Maximum VDC	Maximum VDC
0	26.8	27.9	30.6	30.8
3	26.3	27.4	29.9	30.0
10	26.2	27.3	29.8	29.9
20	26.1	27.2	29.7	29.8
50	25.7	26.8	29.3	29.4
100	25.2	26.3	28.8	28.9
150	25.0	26.1	28.6	28.7
200	24.7	25.7	28.2	28.3
250	24.5	25.5	28.0	28.1
300	24.0	25.0	27.5	27.6

FIGURE 3. Output voltage limits.

GENERAL NOTES:

This converter shall conform to the:

(1) dimensions and configurations specified on Figure 1 of this specification sheet.

(2) connection diagram specified on Figure 2 of this specification sheet.

(3) converter characteristics specified in Table I of this specification sheet.

(4) output voltage limits specified on Figure 3 of this specification sheet.

REQUIREMENTS:

The converter shall meet the requirements of MIL-PRF-7115 except as modified below (paragraph number refers to MIL-PRF-7115). (Reference MIL-PRF-7115 paragraph 3.1.)

1.2.1 Classification. The converter will be classified as a Type II non-regulated converter designed for a nominal 28VDC output.

1.2.2 Converter Class. Delete and add: The converter will be classified as a class A with internal cooling fan.

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3.4.2.1.1.1 Terminals block design. Add as second sentence: Output studs, as shown on Figure 1 of this specification sheet, are to be of different stud sizes to preclude the inadvertent reverse polarity connection of the TRU during aircraft installation.

3.4.3.7 Size and weight. Delete and add: The dimensions and weight of the converter shall be as shown as on Figure 1 of this specification sheet.

3.5.1.1.7.1 Input current waveform. Add to first sentence: With exception of the 5th, 7th, 11th and 13th harmonic, which shall not cause harmonic line currents greater the 6 percent of the converters full rated load fundamental current.

3.5.1.2.1 Steady state output voltage. Add as first sentence: The output voltage shall remain within the limits on Figure 3 of this specification sheet.

3.5.1.2.1.1c Peak to Peak ripple voltage. Delete and add: The ripple voltage shall not exceed 1.5 volts (peak-peak) from zero to 100 percent load current.

3.5.1.5 Acoustical noise. Delete and add: Sound pressure levels of the converter shall not exceed those listed in Table II of this specification sheet.

3.6 Reliability. Add as second sentence: Analysis to be done in accordance with 6.4.4 of MIL-PRF-7115 and results shall be provided to qualifying activity.

3.7.4 Airflow. Add as third sentence: The direction of airflow shall be as shown on Figure 1 of this specification sheet and depicted by an arrow that is visible on the converter exterior.

4.6.2.12 Overload and short circuit capacity. Delete second/third sentence and add: Tests shall be conducted at the extremes of input voltage and frequency in 3.5.1.1 and 3.5.1.1.1.1. Tests on Type II converters shall be conducted at the test points indicated on Figure 9 from 120 percent through 300 percent.

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents:
AIA/NASM315 AIA/NAS1149 SAE-AS34521 SAE-AS34561

Custodian:
Navy - AS

Preparing activity:
Navy - AS
(Project 6130-2014-004)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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